

Muhammad Talal Akhtar

Computer Science Student | Aspiring Software Engineer

muhammadtalalakhtar01@gmail.com | +92 332 6516291 | Islamabad, Pakistan

[GitHub Profile](#)

SUMMARY

Computer Science student at FAST-NUCES with a strong focus on full-stack development. Experienced in building scalable MERN applications with RESTful APIs, JWT-based authentication, and efficient data modeling. Solid foundation in data structures and algorithms, applied through system-level C++ projects. Focused on writing clean, modular code and solving real-world problems through practical, production-oriented projects.

TECHNICAL SKILLS

Languages: C++ (Advanced), JavaScript, Python

Frontend: React.js, HTML5, CSS3, Responsive Design

Backend: Node.js, Express.js, REST API Design, JWT Authentication, bcrypt

Databases: MongoDB + Mongoose

Core Concepts: Data Structures & Algorithms, OOP, Graph Theory, Stack/Queue, Modular System Design

Tools: Git, GitHub, VS Code, Postman

Practices: Debugging, Performance Optimization, Code Reviews

PROJECTS

Personal Finance Tracker

[GitHub ↗](#)

Node.js | Express.js | React.js | MongoDB | JWT | REST APIs

- Architected a full-stack MERN application with secure JWT + bcrypt authentication, ensuring protected access to all users.
- Developed 12+ RESTful API endpoints for transaction management with structured JSON responses and centralized error handling.
- Designed MongoDB schemas with Mongoose validation, enforcing data integrity and reducing invalid writes.
- Optimized API communication to reduce redundant requests and improve application responsiveness.

Advanced Social Network System

[GitHub ↗](#)

C++ | Data Structures | Graph Algorithms | OOP

- Designed a terminal-based backend system modeling user relationships using weighted graphs based on interaction metrics.
- Implemented BFS and DFS algorithms for relationship traversal and friendship discovery features.
- Applied modular OOP principles with decoupled classes, improving scalability and code maintainability.
- Demonstrated strong understanding of graph theory and system-level problem solving in C++.

Personal Portfolio Website

[Live ↗](#)

HTML5 | CSS3 | JavaScript | Responsive Design

- Developed a fully responsive, mobile-first portfolio website with optimized performance.
- Achieved fast load times through lightweight design and efficient asset handling.
- Implemented interactive UI elements and smooth navigation using vanilla JavaScript.

EDUCATION

National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad

Bachelor of Science in Computer Science